

CSE200: Online-1 on L^AT_EX

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1 Introduction

Document preparation systems are essential for producing structured and professional technical documents. **L^AT_EX** allows authors to focus on content while maintaining consistent formatting. This article demonstrates formatted text, lists, equations, tables, and figures using commands taught in class.

1.1 Text Emphasis and Fonts

This sentence contains **bold text**, *italic text*, *emphasized text*, and underlined text. Font sizes can also vary such as Large text, small text, and huge text.

2 List Structures

2.1 Mixed and Nested Lists

- Primary Feature
- Secondary Feature
 1. Formatting control
 2. Mathematical typesetting
 - Inline math
 - Display math

Note Lists can be customized

3 Mathematical Modeling

Mathematical expressions are often used to represent data relationships. Consider variables x_i , y and z^2 .

4 Mathematical Modeling

Mathematical expressions are commonly used to describe relationships among variables. Let the variables be x_i , y_j and σ^2 .

4.1 Displayed Equations

$$y_j = \alpha x_j^2 + \beta x_j + \gamma \tag{1}$$

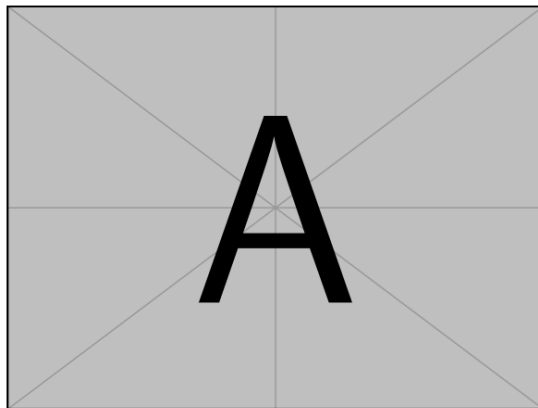
$$\begin{aligned} T &= \sum_{i=1}^n (x_i - \mu)^2 \\ &= \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2 \end{aligned} \tag{2}$$

$$f(x_i) = \begin{cases} \frac{x_i^2}{\sigma^2} & \text{if } x_i \geq 0 \\ \frac{|x_i|}{\sigma} & \text{if } x_i < 0 \end{cases}$$

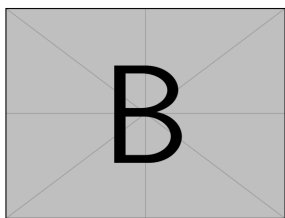
5 Tabular Data Presentation

6 Visual Comparison

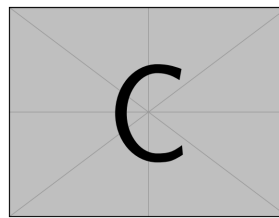
6.1 Asymmetric Subfigure Layout



(a) Main Diagram



(b) Detail B



(c) Detail C

Figure 1: Asymmetric layout with one dominant figure.

Figure 1 illustrates how visual elements can be arranged systematically within a document.

7 Conclusion

This article highlighted the importance of **structured writing** using $\text{\LaTeX}$. The use of *formatted text*, mathematical expressions, tables, and figures improves both readability and presentation quality.